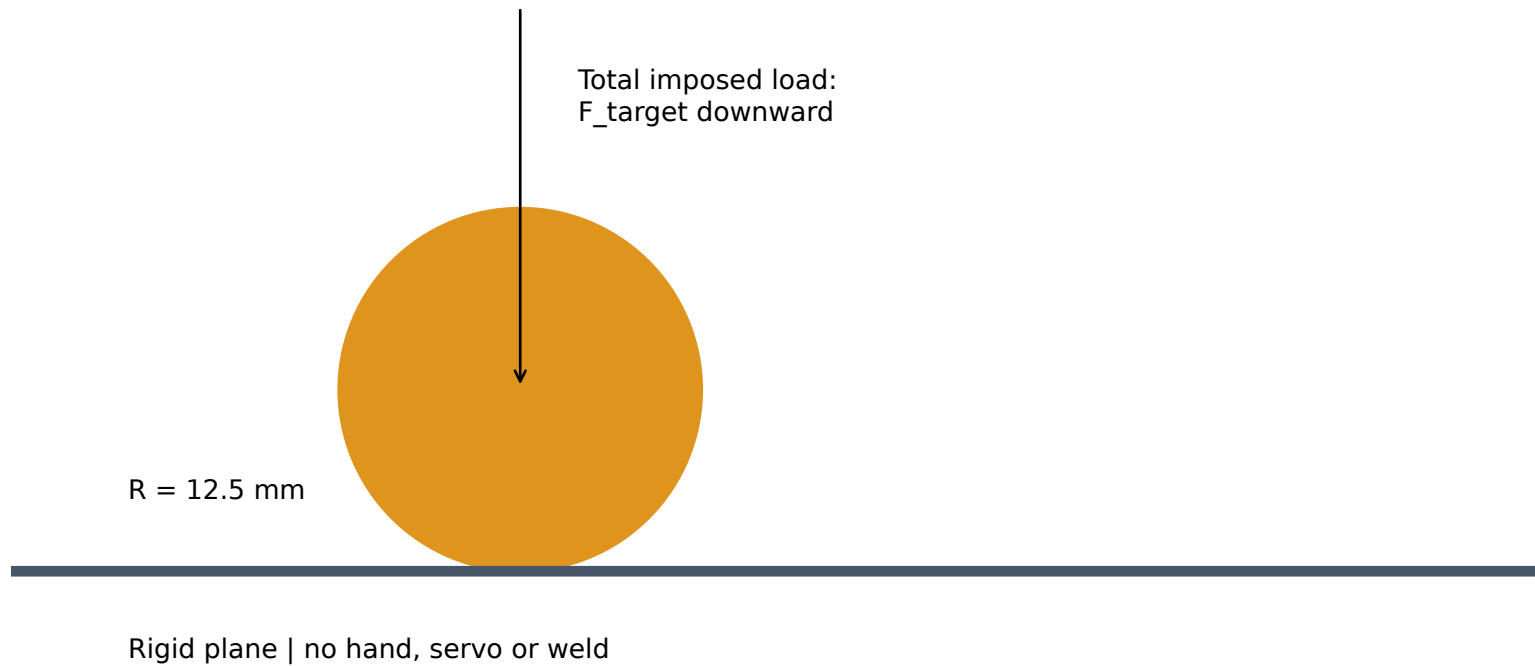


Isolated normal-contact bench: total load is prescribed



Gravity retained. Applied COM force is $mg - F_{\text{target}}$; sphere starts tangent with zero velocity.